

Sam Butler

sambutle@andrew.cmu.edu • [Website](#) • [LinkedIn](#) • [Github](#) • 267-586-0022 • he/him

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Master of Science in Applied Data Science

Expected May 2027

Relevant Courses: Statistical Computing, Applied Linear Models, Data Engineering/Visualization

Pomona College, Claremont, CA

Bachelor of Arts; GPA 3.83/4.00

Graduated May 2026

Major: Math - Statistics

Athletics: Varsity Track & Field Captain

TECHNICAL SKILLS

Programming Languages: R (base and Tidyverse), co-author of R package Sicegar, currently learning Python, SQL

Optimization: Both traditional and stochastic methods (gradient descent, metaheuristic algorithms)

Statistical Modeling: Machine Learning, Regression, Classification, Random Forests, gradient boosting

Data Analysis: Cleaning/Tidying data, Data Visualization (ggplot2, Shiny, plotly), Data collection

Mentorship/Teaching: Have taught middle school math, tutored, TA'd multiple stats courses, led backpacking trips

EXPERIENCE

Pomona College, Claremont, CA

May 2025 - February 2026

Research Assistant

- Rewrote and debugged R package (Sicegar) to improve accuracy and reliability of nonlinear curve-fitting methods
- Developed data visualizations that synthesized complex model outputs into clean, interpretable insights
- Accurately and completely documented simulation procedures and findings (using GitHub as version control)
- Presented weekly research updates to faculty and peers, communicating technical results to diverse audiences
- Co-authored a paper on revised Sicegar methodology; manuscript submitted to the *Journal of Open Source Software*

Pomona College, Claremont, CA

September 2025 - May 2026

Stats Mentor/TA

- Served as a TA for statistics courses, supporting students through weekly office hours and problem-solving sessions
- Translated statistical concepts into clear, intuitive explanations for students with diverse quantitative backgrounds

Apogee Adventures, Brunswick, ME

Summer 2024, Summer 2026

Outdoor Trip Guide

- Led and supervised groups of 12-16 year olds on multi-week wilderness trips, including 105 mile Tour du Mont Blanc
- Collaborated with fellow leaders to prepare and manage trip logistics, including gear, food supplies, and transportation
- Obtained Wilderness First Aid, Wilderness Swim Safety, and CPR Certification

Breakthrough Collaborative, Philadelphia, PA

May 2023 - August 2023

Teaching Fellow

- Designed and implemented Pre-Algebra curriculum, tailored to satisfy the standards of The School District of Philadelphia
- Delivered interactive lessons to a diverse group of learners fostering critical thinking and problem-solving skills
- Practiced community stewardship by mentoring historically underrepresented students in Philadelphia

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PERSONAL PROJECTS

- **Thesis:** Researched and created algorithm hybridizing the Levenberg-Marquardt and Whale Optimization Algorithms to better optimize multimodal search spaces
- **University Analytics:** Built interactive Shiny app predicting university yield rates via XGBoost gradient boosting and segmenting institutions into peer clusters using K-Means and Random Forest-based cluster characterization
- **Tidy Tuesday:** Designed interactive R visualizations analyzing solar eclipse paths, volcanic seismic activity, and roundabout infrastructure, using statistical packages (sf, ggspatial) for geospatial mapping

SKILLS & INTERESTS

French Fluency - Reading, Writing, Speaking